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(57) Abstract :

People who are deaf or hard of hearing use sign languages to communicate with each other through hand, facial, and body gestures. Hand gestures are one of their primary means of communication because, despite the variety of methods accessible, no suitable technologies exist to facilitate linking this social group with the outside world. Machine learning and image classification can be used to assist computers in recognizing sign language, which can subsequently be deciphered by others. Learning the finger sign alphabets is the fundamental method of learning sign language. This product primarily concentrates on learning alphabetic sign languages, which serves as the foundational stage for learning hand movements. The majority of current technologies employ electromyography techniques, CNN algorithms, and machine learning algorithms, although they are not as accurate as the Yolo approach. This method offers 95.7% accuracy. This primary goal of this work is to use the Yolo model to accurately identify the gesture.

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